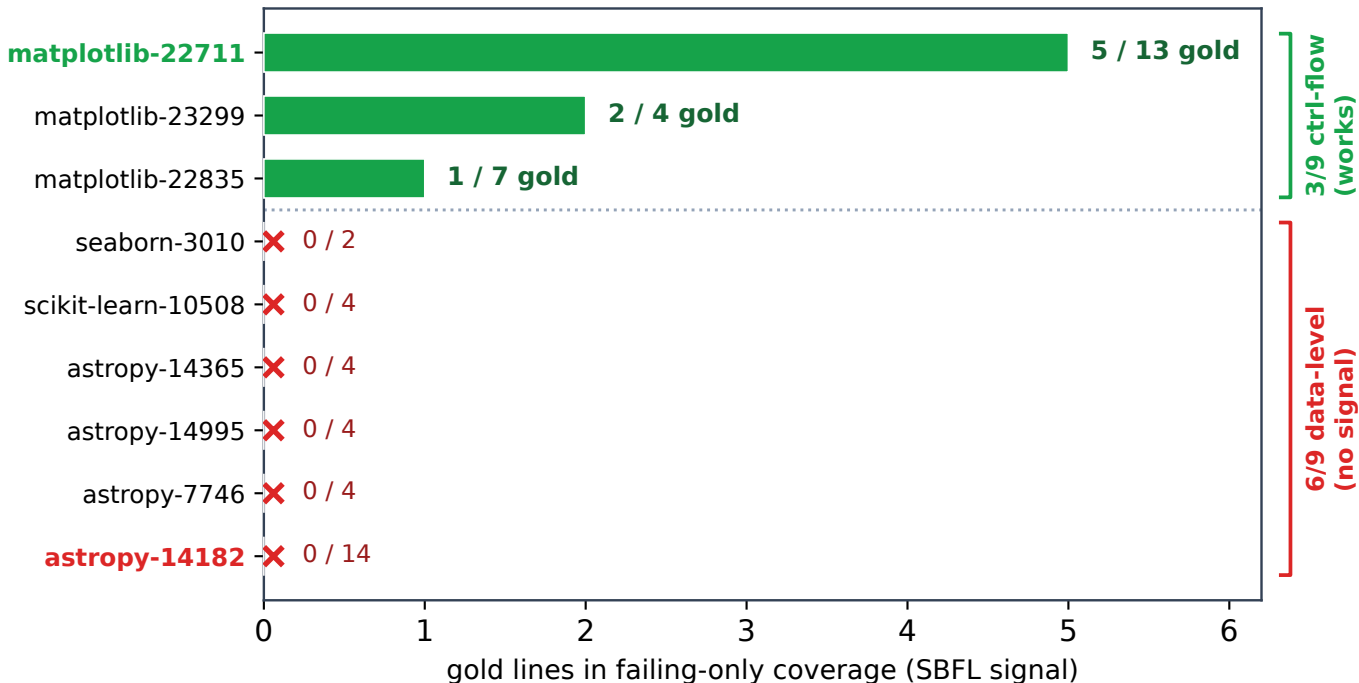


RQ2.3

The coverage differential does not rank

Necessary condition tested: gold is in the failing\passing spectrum, so SBFL floats it up.



X the failing-passing differential is empty on 6/9 — it cannot rank the fault

Focal — astropy-14182

failing cov = 25 lines
passing cov = 28 lines
failing $\subseteq$ passing

$\Rightarrow$ fail-only = $\emptyset$, gold 0/14
 $\Rightarrow$ SBFL has nothing to rank.

Contrast matplotlib-22711:
fail-only 93 lines, gold 5/13
 $\rightarrow$ the signal works.

Root cause

Data-level faults run under BOTH failing and passing tests $\rightarrow$ empty differential; it fires only for the control-flow minority (3/9).